

Lightweight Linux GUI Distro for Small-Scale Private Cloud Deployment

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Abstract—The increasing adoption of private cloud environments has created demand for lightweight operating systems capable of delivering cloud services while maintaining low resource consumption. This study evaluates three lightweight Linux graphical user interface (GUI) distributions, namely Ubuntu, Xubuntu, and Linux Lite, to determine their suitability for small-scale private cloud deployment. The distributions were installed and tested within identical VMware virtual machine environments. Baseline performance metrics including memory consumption, boot times, and storage utilisation were collected. Docker and a nginx container were subsequently deployed to simulate a lightweight cloud service. Results indicate that Linux Lite demonstrated the lowest memory consumption during both idle and workload conditions, while Ubuntu achieved the fastest startup performance. All tested distributions successfully hosted containerised services without stability issues. The findings suggest that lightweight Linux distributions can provide practical and cost-effective solutions for small-scale private cloud environments.

Keywords—Linux, private cloud, Docker, virtualization, Ubuntu, Xubuntu, Linux Lite, lightweight operating systems.

I. INTRODUCTION

Private cloud computing has become an increasingly popular solution for individuals, educational institutions, and small organisations seeking greater control over their data and services [1]. Unlike public cloud platforms, private clouds operate within dedicated environments, providing enhanced privacy, security, and flexibility [2]. Small-scale private cloud deployments are commonly used for web hosting, file sharing, backup services, and personal infrastructure management. However, many users deploying private cloud solutions face hardware limitations, making resource-efficient operating systems an important consideration [3].

Linux-based operating systems are widely utilised in cloud environments due to their stability, open-source nature, and extensive community support [4]. While server-oriented Linux distributions are commonly used in large-scale deployments, lightweight Linux graphical user interface (GUI) distributions offer an attractive alternative for smaller environments where ease of use and low resource consumption are equally important [5]. These distributions provide desktop functionality while maintaining relatively low hardware requirements, making them suitable candidates for resource-constrained cloud systems.

Several lightweight Linux distributions are available, each employing different desktop environments and system optimisations. Ubuntu uses the Lightweight Qt (LXQt) desktop environment, which is designed to minimise resource consumption while maintaining usability. Xubuntu and Linux Lite utilise the XFCE desktop environment, offering a balance between performance and user experience [5]. Despite their popularity, there is limited practical comparison of these distributions in the context of small-scale private cloud deployment.

This paper examines the suitability of Ubuntu, Xubuntu, and Linux Lite for lightweight private cloud environments. The study evaluates their performance through virtualised testing, measuring key metrics such as memory usage, startup performance, resource consumption under workload, and overall system responsiveness. Docker and a nginx container are utilised to simulate a lightweight cloud service. The findings aim to identify which distribution provides the most effective balance between resource efficiency, usability, and functionality for small-scale private cloud deployment.

II. RELATED WORK

Lightweight Linux distributions have long been recognised as effective solutions for systems with limited hardware resources. Their reduced memory footprint and lower processing requirements make them suitable for older hardware, embedded systems, and virtualised environments. Previous studies have shown that the choice of desktop environment can significantly influence overall system performance, particularly in terms of memory consumption, startup times, and responsiveness. Desktop environments such as LXQt and XFCE are commonly used in lightweight distributions because they provide graphical functionality while maintaining relatively low resource usage [6].

Ubuntu, Xubuntu, and Linux Lite are among the most widely adopted lightweight Linux distributions. Ubuntu utilises the LXQt desktop environment and is often recommended for low-specification hardware due to its minimal resource requirements. Xubuntu employs the XFCE desktop environment, offering additional features and usability enhancements while maintaining moderate resource consumption. Linux Lite is also based on XFCE but focuses on providing a familiar user experience for users transitioning from Microsoft Windows [6]. Although these distributions are frequently recommended for resource-constrained

systems, comparative evaluations within private cloud environments remain limited.

In parallel with developments in lightweight operating systems, containerisation technologies have become increasingly important in cloud computing. Docker has emerged as one of the most widely used container platforms due to its ability to package applications and their dependencies into isolated environments. Compared with traditional virtual machines, containers generally require fewer system resources and provide faster deployment times [7]. This efficiency makes Docker particularly suitable for small-scale private cloud infrastructures where hardware resources may be limited.

The combination of lightweight Linux distributions and containerised services offers a practical approach to resource-efficient cloud deployment. However, relatively few studies have examined how different lightweight Linux GUI distributions perform when hosting containerised workloads. This research contributes to addressing that gap by comparing Lubuntu, Xubuntu, and Linux Lite under identical virtualised conditions and evaluating their suitability for small-scale private cloud deployment.

III. METHODOLOGY

A comparative experimental approach was adopted to evaluate the suitability of lightweight Linux GUI distributions for small-scale private cloud deployment. Three operating systems were selected for testing: Lubuntu, Xubuntu, and Linux Lite. These distributions were chosen because of their reputation for low resource consumption while still providing fully functional graphical desktop environments.

To ensure consistency and fairness throughout the evaluation, all testing was conducted within VMware using identical virtual machine configurations. Each virtual machine was allocated 4 GB of RAM, 2 virtual CPU cores, and 20 GB of virtual storage. Network connectivity was configured using Network Address Translation (NAT) to provide internet access for software installation and system updates. The operating systems were installed using the “Erase disk and install” option to maintain a consistent storage configuration across all virtual machines.

Following installation, each distribution was updated using the Advanced Package Tool (APT) package management system. System monitoring and benchmarking tools were then installed to collect performance data. These included htop for monitoring CPU and memory utilisation, Fastfetch for displaying system information, systemd-analyze for measuring startup performance, and df -h for assessing disk usage.

```
neenak@neena-vmwarevirtualplatform:~$ fastfetch
neenak@neena-vmwarevirtualplatform:~$
OS: Ubuntu 26.04 LTS (Resolute 4)
Host: VMware Virtual Platform
Kernel: Linux 7.0.0-15-generic
Uptime: 27 mins
Packages: 1974 (dpkg), 8 (snap)
Shell: bash 5.3.9
Display (Virtual-1): 1280x800, z
DE: LXQt
WM: Openbox (X11)
Theme: Lubuntu [Qt], Breeze [GTK]
Icons: Papyrus [Qt], Papyrus [GTK]
Font: Ubuntu (11pt) [Qt], Ubuntu
Cursor: breeze (18px)
Terminal: qterminal 2.3.0
Terminal Font: Ubuntu Mono (14p)
CPU: 2 x Intel(R) Core(TM) i7-82
GPU: VMware SVGA II Adapter
Memory: 962.43 MiB / 3.77 GiB ( )
Swap: 0 B / 512.00 MiB (0%)
Disk (/): 8.32 GiB / 19.52 GiB 4
Disk (/run/media/neenak/Lubuntu)
Local IP (ens33): 192.168.104.14
Locale: en_GB.UTF-8
```

Fig. 1. Lubuntu system information obtained using Fastfetch

The first stage of testing involved collecting baseline performance measurements. Each operating system was allowed to remain idle for several minutes after startup before measurements were recorded. Metrics collected during this stage included idle RAM usage, startup time, and storage utilisation. These measurements provided an indication of the resource requirements of each distribution under normal operating conditions.

The second stage focused on private cloud deployment. Docker was installed on each distribution and used to deploy a nginx container. The nginx service was selected because it provides a lightweight and widely used web server platform suitable for demonstrating private cloud functionality [8]. Successful deployment was verified by accessing the default nginx web page through a web browser running in each virtual machine.

Following deployment, additional performance measurements were collected while the containerised service was running. Resource usage was monitored using htop and free -m, with particular attention given to memory consumption and CPU utilisation. Observations regarding system responsiveness, usability, and stability were also recorded throughout the testing process. The resulting data was analysed and compared to determine which distribution provided the most suitable balance between resource efficiency, usability, and functionality for lightweight private cloud deployment.

```
0[| 3.3%] Tasks: 111, 485 thr, 187 kthr; 1 runni
1[| 0.7%] Load average: 0.00 0.09 0.14
Mem[| 1.23G/3.77G] Uptime: 00:15:47
Swp[| 0K/3.02G]

Main I/O
PID USER PRI NI VIRT RES SHR S CPU%MEM+ TIME+ Command
5097 neema-kuli 20 0 11820 5612 3932 R 3.3 0.1 0:00.55 htop
706 root 19 -1 59100 27676 26088 S 0.7 0.7 0:00.41 /usr/lib/syst
1039 root 20 0 113M 10900 9256 S 0.7 0.3 0:02.33 /usr/bin/vmto
4591 neema-kuli 20 0 2613M 155M 89072 S 0.7 4.0 0:03.03 /snap/firefox
1 root 20 0 25144 16468 11744 S 0.0 0.4 0:03.59 /usr/lib/syst
735 root 20 0 155M 1584 1276 S 0.0 0.0 0:00.00 vmware-vmblc
736 root 20 0 155M 1584 0 S 0.0 0.0 0:00.00 vmware-vmblc
755 systemd-re 20 0 22936 15128 12040 S 0.0 0.4 0:00.29 /usr/lib/syst
759 root 20 0 44840 18272 9232 S 0.0 0.5 0:00.81 /usr/lib/syst
1026 root 20 0 23156 11356 9916 S 0.0 0.3 0:00.02 /usr/bin/VGAu
1125 root 20 0 113M 10900 0 S 0.0 0.3 0:00.00 /usr/bin/vmto
1133 root 20 0 304M 7992 7220 S 0.0 0.2 0:00.09 /usr/libexec/
1135 root 20 0 113M 10900 0 S 0.0 0.3 0:00.09 /usr/bin/vmto
1137 root 20 0 113M 10900 0 S 0.0 0.3 0:00.00 /usr/bin/vmto
1138 avahi 20 0 6684 4764 4368 S 0.0 0.1 0:00.13 avahi-daemon:
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice F9Kill F10Quit
```

Fig. 2. Xubuntu system resource monitoring using htop during Docker/nginx operation.

IV. IMPLEMENTATION

The implementation phase focused on configuring each Linux distribution and deploying a lightweight private cloud service within the virtualised environment. After completing system updates, Docker was installed on Ubuntu, Xubuntu, and Linux Lite using the standard Ubuntu package management process [9]. Due to the Ubuntu-based architecture of all three distributions, the installation procedure remained largely identical across the test environments and no significant compatibility issues were encountered.

Following the Docker installation, a nginx container was deployed on each virtual machine to simulate a lightweight cloud-based service.

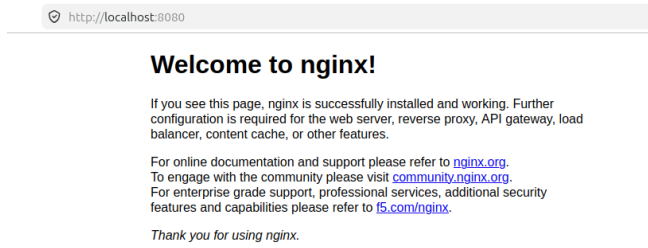


Fig. 3. Successful nginx container deployment within the virtualized environment.

The container was configured to provide a basic web server accessible through a web browser. Successful deployment was verified by accessing the default nginx web page, which confirmed that Docker was functioning correctly and that the containerised service was operational.

During implementation, screenshots were collected as evidence of practical work. This included system information obtained through Fastfetch, resource monitoring data collected using htop, and the nginx web server running within each virtual machine. The screenshots served as supporting evidence for the deployment process and provided visual confirmation of system functionality.

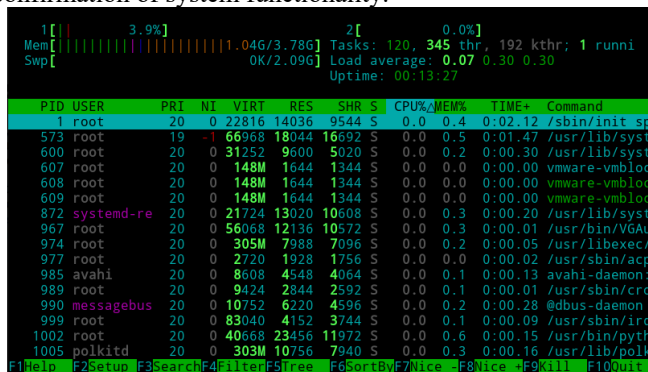


Fig. 4. *Linux Lite system monitoring during workload testing*

Once the cloud service was operational, resource monitoring tools were used to observe system behaviour under workload conditions. Memory usage and CPU utilisation were recorded while the nginx container remained active. Attention was given to system responsiveness and overall stability to determine whether the distributions could reliably support containerised services within a small-scale private cloud environment.

The successful deployment and operation of Docker and nginx across all three distributions demonstrated that each operating system could host lightweight cloud services. This implementation stage provided the practical foundation for the performance evaluation and comparative analysis presented in the following section.

Additional screenshots, configuration details, weekly reports, and supporting materials required to replicate the experimental environment are available in the accompanying GitHub repository [10].

V.RESULTS & ANALYSIS

The performance evaluation compared Ubuntu, Xubuntu, and Linux Lite under both idle conditions and during the execution of a containerised nginx service. The collected measurements provided insight into the resource efficiency, responsiveness, and overall suitability of each distribution for small-scale private cloud deployment.

Table I presents the baseline performance results obtained after each operating system was allowed to remain idle following startup. Linux Lite recorded the lowest idle memory usage at 892 MB, while Ubuntu and Xubuntu recorded 958 MB and 998 MB respectively. These findings suggest that Linux Lite maintained the smallest memory footprint during normal operation. Startup performance measurements showed a different trend, with Ubuntu achieving the fastest boot time of 13.047 seconds. Xubuntu followed with a startup time of 15.610 seconds, while Linux Lite required 42.072 seconds to reach a fully operational state.

A. Baseline Performance

TABLE I BASELINE PERFORMANCE RESULTS

Distribution	Idle RAM Usage (MB)	Boot Time (s)
Lubuntu	958	13.047
Xubuntu	998	15.610
Linux Lite	892	42.072

Following the deployment of Docker and the nginx container, additional measurements were collected to assess performance under workload conditions. All three distributions successfully hosted the containerised service without errors or service interruptions. Resource monitoring indicated that Linux Lite continued to demonstrate the lowest memory consumption, using approximately 1313 MB of RAM while the nginx service was running. Xubuntu consumed approximately 1402 MB, while Ubuntu recorded the highest memory usage at 1437 MB.

B. Docker Workload Performance

TABLE II RESOURCE USAGE DURING DOCKER/NGINX DEPLOYMENT

Distribution	RAM Usage Under Load (MB)
Lubuntu	1437
Xubuntu	1402
Linux Lite	1313

CPU utilisation remained relatively low throughout testing, generally fluctuating between 1% and 4% across all distributions. This indicates that the nginx workload placed

minimal processing demands on the virtual machines and that memory consumption represented the more significant differentiating factor between the operating systems. No system crashes, service failures, or stability issues were observed during the workload testing phase.

C. Discussion

In addition to the quantitative measurements, usability observations were recorded throughout the evaluation process. Lubuntu provided a highly responsive desktop environment due to its lightweight LXQt interface and demonstrated the fastest startup performance. Xubuntu delivered a more feature-rich user experience through the XFCE desktop environment but generally consumed slightly more resources during idle operation. Linux Lite offered a familiar and accessible interface while consistently demonstrating the lowest memory consumption both before and after deployment of the containerised service.

Overall, the results indicate that all three distributions can support lightweight private cloud workloads. However, Linux Lite achieved the best memory efficiency, while Lubuntu provided superior startup performance. These findings suggest that the choice of distribution may depend on whether resource efficiency or startup responsiveness is considered the primary deployment requirement.

VI. CONCLUSION

This study evaluated the suitability of three lightweight Linux GUI distributions (Lubuntu, Xubuntu, and Linux Lite) for small-scale private cloud deployment. The distributions were tested within identical virtualised environments and assessed using baseline performance measurements and a Docker-based nginx deployment. Key metrics included memory usage, startup performance, resource consumption under workload, and overall system responsiveness.

The results demonstrated that all three distributions were capable of successfully hosting a lightweight containerised service without stability issues. Linux Lite consistently recorded the lowest memory consumption during both idle and workload conditions, making it the most resource-efficient option among the tested systems. Lubuntu achieved the fastest startup performance and provided a highly responsive user experience, while Xubuntu offered a more feature-rich desktop environment at the cost of slightly higher resource usage.

Overall, the findings indicate that lightweight Linux distributions provide a practical and cost-effective platform for small-scale private cloud environments. The selection of a specific distribution ultimately depends on deployment priorities, with Linux Lite favouring memory efficiency and Lubuntu offering superior startup responsiveness. Supporting documentation, source files and additional practical evidence have been made publicly available through a GitHub repository to facilitate reproducibility of the work. Future work could expand the evaluation by testing additional cloud services, larger workloads, and deployment on physical hardware rather than virtual machines.

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